# The Brain as a Matrix

Modern neuroscience often begins with one simple question: **How does the brain do X?**

- When a person thinks, hears, or speaks, billions of neurons communicate through tiny electrical signals.
- In imaging neurosciences, we try to answer this question by recording brain activity while human experimental subjects perform a set of cognitive tasks.
- With imaging tools such as **magnetoencephalography (MEG)**, sensors placed around the head measure these signals from many brain regions at once. This gives us a huge amount of data

**How do we make sense of large, complex data from the human brain?**

The result is a rich dataset:

- **3D coordinates** describing where each brain region is located, and
- **time-series activity** showing how active each region is over time.

Both of these datasets are high-dimensional,and difficult to interpret directly.

**Linear algebra provides the language we need to analyze, compare, and understand them. and MATLAB offers the computational tools to implement these concepts efficiently.**

In this assignment you will apply linear algebra on neuroimaging data going step-by-step from spatial transformations, to dimensionality reduction, to network analysis, mirroring the same computational pipeline used in modern neuroimaging research.

To understand the structure hidden in this large matrix, neuroscientists rely on linear algebra tools that reveal patterns such as:

- Which brain regions activate together
- How strongly regions influence one another
- Which regions are the most connected or “important” in the network

We can think of the data recorded from the brain as a large data matrix

$$S \in \mathbb{R}^{T \times R}$$

where:

- $T$ = number of time points
- $R$ = number of brain regions

Then we'll apply the techniques we learned in linear algebra to make sense of this neuroimaging data.

In this assignment, you will use our newly acquired linear algebra skills to process brain data for an MEG experiment.

1. **Load data into MATLAB**
2. **3D coregistration**: Transforming recording locations to account for head movement inside the brain scanner
3. **Dimensionality reduction (PCA):** Find groups of brain regions whose activity varies together. Instead of looking at all 100 signals individually, PCA summarizes them into a few components that capture the main patterns of variation.
4. **Correlation with neurocognitive assessment scores**

*Pro-tip:* All the operations you need for this assignment are provided in the MATLAB exercises (Quercus), and MATLAB Academy On-Ramp and Linear Algebra learning modules

## Problem 1 - Load neuroimaging data into MATLAB

➡ First, **load the subject's brain coordinates from CSV** using `readtable` from https://github.com/dtxe/mat188_datasets/raw/refs/heads/main/schaefer/Subject200_Centroids_linear.csv

```
% Clear our workspace to prevent interference from previously loaded data
clear

% Load the subject's coordinates
```

➡ Create matrix $L$, with dimensions n x 3 where n is the number of coordinates.

```
coords_tb1 = readtable('https://raw.githubusercontent.com/dtxe/
mat188_datasets/refs/heads/main/schaefer/Subject200_Centroids_linear.csv');
L = table2array(coords_tb1)
```

```
L = 200×3
-36.7718    -50.9494    -10.2000
-44.0120    -73.6969    -14.2800
-63.3052    -65.8682    -8.1600
-25.0654    -64.3995    -4.0800
-47.8667    -90.7753    -12.2400
-24.1271    -41.5805    -2.0400
-26.0036    -87.2184    -4.0800
-65.4059    -65.8444    10.2000
-43.6654    -90.8230    6.1200
-27.5943    -66.2733    8.1600
-60.0014    -79.2229    10.2000
-28.0226    -68.1709    22.4400
-25.1470    -83.4232    28.5600
-44.0528    -83.2088    24.4800
-53.3727    -3.1995     -2.0400
```

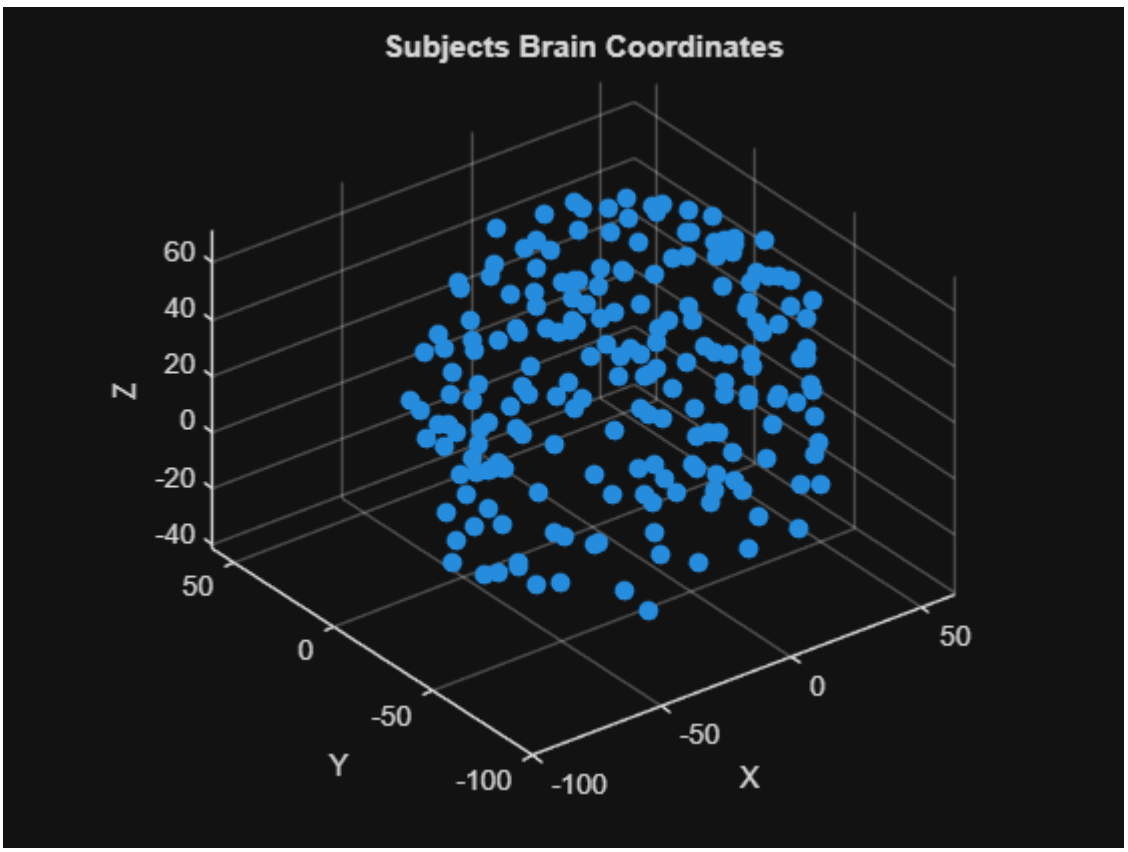
⋮

**Plot the subjects's brain coordinates in 3D**

➡ **Plot** the subjects's brain coordinates as a 3D scatter plot

- Remember to use `daspect([1, 1, 1])` so that MATLAB keeps the same scale for x, y, and z axes

```
figure;
scatter3(L(:,1), L(:,2), L(:,3), 50, 'filled');
xlabel('X'); ylabel('Y'); zlabel('Z');
title('Subjects Brain Coordinates');
daspect([1 1 1]);
grid on;
```




**Load brain activity dataset**

Next, we are given a matrix of brain activity recorded using MEG:

$$S \in \mathbb{R}^{T \times R}$$

where:

- $T$ = number of time points
- $R$ = number of brain regions

In other words, each column of $S$ is the activity of one brain region over time.

### Load the data

➡ **Load an MEG recording** from an individual from `https://github.com/dtxe/mat188_datasets/raw/refs/heads/main/meg/megdata.csv`

- Hint: Use the `readtable` function, that can read CSV files directly from a URL

```
meg_table = readtable('https://github.com/dtxe/mat188_datasets/raw/refs/heads/main/meg/megdata.csv');
```

➡ Inspect the table using the `summary()` function and by opening it in the Workspace.

```
summary(meg_table)
```

```
meg_table: 500×201 table

Variables:

    Time: double
    LH_VisCent_ExStr_1: double
    LH_VisCent_ExStr_2: double
    LH_VisCent_Striate_1: double
    LH_VisCent_ExStr_3: double
    LH_VisCent_ExStr_4: double
    LH_VisCent_ExStr_5: double
    LH_VisPeri_ExStrInf_1: double
    LH_VisPeri_ExStrInf_2: double
    LH_VisPeri_ExStrInf_3: double
    LH_VisPeri_StriCal_1: double
    LH_VisPeri_ExStrSup_1: double
    LH_VisPeri_ExStrSup_2: double
    LH_SomMotA_1: double
    LH_SomMotA_2: double
    LH_SomMotA_3: double
    LH_SomMotA_4: double
    LH_SomMotA_5: double
    LH_SomMotA_6: double
    LH_SomMotA_7: double
    LH_SomMotA_8: double
    LH_SomMotB_Aud_1: double
    LH_SomMotB_Aud_2: double
    LH_SomMotB_S2_1: double
    LH_SomMotB_S2_2: double
    LH_SomMotB_Aud_3: double
    LH_SomMotB_S2_3: double
    LH_SomMotB_Cent_1: double
    LH_SomMotB_Cent_2: double
```

**LH_DorsAttnA_TempOcc_1**: double
**LH_DorsAttnA_TempOcc_2**: double
**LH_DorsAttnA_ParOcc_1**: double
**LH_DorsAttnA_SPL_1**: double
**LH_DorsAttnA_SPL_2**: double
**LH_DorsAttnA_SPL_3**: double
**LH_DorsAttnB_PostC_1**: double
**LH_DorsAttnB_PostC_2**: double
**LH_DorsAttnB_PostC_3**: double
**LH_DorsAttnB_PostC_4**: double
**LH_DorsAttnB_FEF_1**: double
**LH_SalVentAttnA_ParOper_1**: double
**LH_SalVentAttnA_Ins_1**: double
**LH_SalVentAttnA_FrOper_1**: double
**LH_SalVentAttnA_FrOper_2**: double
**LH_SalVentAttnA_ParMed_1**: double
**LH_SalVentAttnA_FrMed_1**: double
**LH_SalVentAttnA_FrMed_2**: double
**LH_SalVentAttnB_IPL_1**: double
**LH_SalVentAttnB_PFCl_1**: double
**LH_SalVentAttnB_Ins_1**: double
**LH_SalVentAttnB_PFCmp_1**: double
**LH_LimbicB_OFC_1**: double
**LH_LimbicB_OFC_2**: double
**LH_LimbicA_TempPole_1**: double
**LH_LimbicA_TempPole_2**: double
**LH_LimbicA_TempPole_3**: double
**LH_LimbicA_TempPole_4**: double
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**LH_ContA_IPS_1**: double
**LH_ContA_IPS_2**: double
**LH_ContA_IPS_3**: double
**LH_ContA_PFCd_1**: double
**LH_ContA_PFClv_1**: double
**LH_ContA_PFCl_1**: double
**LH_ContA_PFCl_2**: double
**LH_ContA_PFCl_3**: double
**LH_ContA_Cingm_1**: double
**LH_ContB_Temp_1**: double
**LH_ContB_IPL_1**: double
**LH_ContB_PFCl_1**: double
**LH_ContB_PFClv_1**: double
**LH_ContB_PFClv_2**: double
**LH_ContC_pCun_1**: double
**LH_ContC_pCun_2**: double
**LH_ContC_Cingp_1**: double
**LH_DefaultA_IPL_1**: double
**LH_DefaultA_PFCd_1**: double
**LH_DefaultA_pCunPCC_1**: double
**LH_DefaultA_pCunPCC_2**: double
**LH_DefaultA_pCunPCC_3**: double
**LH_DefaultA_PFCm_1**: double
**LH_DefaultA_PFCm_2**: double
**LH_DefaultA_PFCm_3**: double
**LH_DefaultB_Temp_1**: double
**LH_DefaultB_Temp_2**: double
**LH_DefaultB_Temp_3**: double
**LH_DefaultB_Temp_4**: double
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**LH_DefaultB_PFCd_3**: double
**LH_DefaultB_PFCd_4**: double
**LH_DefaultB_PFCv_1**: double

**LH_DefaultB_PFCv_2**: double
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**LH_DefaultB_PFCv_4**: double
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**LH_DefaultC_Rsp_1**: double
**LH_DefaultC_PHC_1**: double
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**RH_VisPeri_StriCal_1**: double
**RH_VisPeri_ExStrSup_1**: double
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**RH_SomMotB_S2_2**: double
**RH_SomMotB_S2_3**: double
**RH_SomMotB_S2_4**: double
**RH_SomMotB_Cent_1**: double
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**RH_DorsAttnA_ParOcc_1**: double
**RH_DorsAttnA_SPL_1**: double
**RH_DorsAttnA_SPL_2**: double
**RH_DorsAttnA_SPL_3**: double
**RH_DorsAttnA_SPL_4**: double
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**RH_DorsAttnB_PostC_3**: double
**RH_DorsAttnB_PostC_4**: double
**RH_DorsAttnB_FEF_1**: double
**RH_SalVentAttnA_ParOper_1**: double
**RH_SalVentAttnA_PrC_1**: double
**RH_SalVentAttnA_Ins_1**: double
**RH_SalVentAttnA_Ins_2**: double
**RH_SalVentAttnA_FrOper_1**: double
**RH_SalVentAttnA_FrMed_1**: double
**RH_SalVentAttnA_ParMed_1**: double
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**RH_SalVentAttnA_FrMed_2**: double
**RH_SalVentAttnB_IPL_1**: double
**RH_SalVentAttnB_PFClv_1**: double
**RH_SalVentAttnB_PFCl_1**: double
**RH_SalVentAttnB_Ins_1**: double
**RH_SalVentAttnB_Ins_2**: double
**RH_SalVentAttnB_PFCmp_1**: double

**RH_LimbicB_OFC_1**: double
**RH_LimbicB_OFC_2**: double
**RH_LimbicB_OFC_3**: double
**RH_LimbicB_OFC_4**: double
**RH_LimbicA_TempPole_1**: double
**RH_LimbicA_TempPole_2**: double
**RH_LimbicA_TempPole_3**: double
**RH_LimbicA_TempPole_4**: double
**RH_ContA_IPS_1**: double
**RH_ContA_IPS_2**: double
**RH_ContA_PFCd_1**: double
**RH_ContA_PFCl_1**: double
**RH_ContA_PFCl_2**: double
**RH_ContA_Cingm_1**: double
**RH_ContB_Temp_1**: double
**RH_ContB_Temp_2**: double
**RH_ContB_IPL_1**: double
**RH_ContB_IPL_2**: double
**RH_ContB_PFCld_1**: double
**RH_ContB_PFCld_2**: double
**RH_ContB_PFClv_1**: double
**RH_ContB_PFClv_2**: double
**RH_ContB_PFCmp_1**: double
**RH_ContB_PFCld_3**: double
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**RH_ContC_Cingp_1**: double
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**RH_DefaultA_PFCd_1**: double
**RH_DefaultA_pCunPCC_1**: double
**RH_DefaultA_PFCm_1**: double
**RH_DefaultA_PFCm_2**: double
**RH_DefaultA_PFCm_3**: double
**RH_DefaultB_Temp_1**: double
**RH_DefaultB_AntTemp_1**: double
**RH_DefaultB_PFCd_1**: double
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**RH_DefaultC_IPL_1**: double
**RH_DefaultC_Rsp_1**: double
**RH_DefaultC_PHC_1**: double
**RH_TempPar_1**: double
**RH_TempPar_2**: double
**RH_TempPar_3**: double
**RH_TempPar_4**: double

Statistics for applicable variables:

| | NumMissing | Min | Median | Max | Mean |
|---|---|---|---|---|---|
| **Time** | 0 | 0 | 0.9980 | 1.9960 | 0.9980 |
| **LH_VisCent_ExStr_1** | 0 | -9.7681 | 0.0031 | 8.5286 | 0.0344 |
| **LH_VisCent_ExStr_2** | 0 | -11.0750 | 0.0189 | 8.9443 | 0.0785 |
| **LH_VisCent_Striate_1** | 0 | -10.4100 | 0.1997 | 8.1248 | 0.0243 |
| **LH_VisCent_ExStr_3** | 0 | -10.5630 | 0.1804 | 8.2671 | 0.0348 |
| **LH_VisCent_ExStr_4** | 0 | -11.5980 | 0.2250 | 9.4753 | 0.0481 |
| **LH_VisCent_ExStr_5** | 0 | -11.0680 | 0.1110 | 9.1014 | 0.0294 |
| **LH_VisPeri_ExStrInf_1** | 0 | -11.2410 | 0.1521 | 8.8263 | 0.0343 |
| **LH_VisPeri_ExStrInf_2** | 0 | -10.8520 | 0.1388 | 9.5620 | 0.0421 |
| **LH_VisPeri_ExStrInf_3** | 0 | -10.9590 | 0.1692 | 9.6198 | 0.0437 |
| **LH_VisPeri_StriCal_1** | 0 | -11.6780 | 0.1938 | 9.9091 | 0.0734 |
| **LH_VisPeri_ExStrSup_1** | 0 | -11.3380 | 0.1319 | 9.2368 | 0.0430 |
| **LH_VisPeri_ExStrSup_2** | 0 | -10.5360 | 0.0405 | 8.2726 | 0.0600 |
| **LH_SomMotA_1** | 0 | -11.5230 | 0.1071 | 9.2983 | 0.0477 |
| **LH_SomMotA_2** | 0 | -11.4830 | 0.0793 | 9.2749 | 0.0379 |

| | | | | | |
|---|---|---|---|---|---|
| **LH_SomMotA_3** | 0 | -10.9510 | 0.1296 | 8.2116 | 0.0543 |
| **LH_SomMotA_4** | 0 | -11.6770 | 0.1834 | 8.7770 | 0.0419 |
| **LH_SomMotA_5** | 0 | -9.7490 | 0.0451 | 7.5235 | 0.0342 |
| **LH_SomMotA_6** | 0 | -11.7240 | 0.1416 | 9.3701 | 0.0352 |
| **LH_SomMotA_7** | 0 | -10.5360 | 0.1665 | 8.5233 | 0.0737 |
| **LH_SomMotA_8** | 0 | -11.7430 | 0.2352 | 9.7144 | 0.0513 |
| **LH_SomMotB_Aud_1** | 0 | -10.1260 | 0.2087 | 8.9215 | 0.0564 |
| **LH_SomMotB_Aud_2** | 0 | -11.6870 | 0.1327 | 9.3202 | 0.0606 |
| **LH_SomMotB_S2_1** | 0 | -11.1950 | 0.0615 | 8.4627 | 0.0395 |
| **LH_SomMotB_S2_2** | 0 | -9.1682 | 0.0912 | 7.6756 | 0.0489 |
| **LH_SomMotB_Aud_3** | 0 | -11.1170 | 0.1634 | 9.3459 | 0.0519 |
| **LH_SomMotB_S2_3** | 0 | -11.1380 | 0.1297 | 9.0593 | 0.0461 |
| **LH_SomMotB_Cent_1** | 0 | -11.2350 | 0.1645 | 9.7953 | 0.0521 |
| **LH_SomMotB_Cent_2** | 0 | -10.6450 | 0.0865 | 8.3524 | 0.0495 |
| **LH_DorsAttnA_TempOcc_1** | 0 | -11.6000 | 0.3265 | 9.2889 | 0.0535 |
| **LH_DorsAttnA_TempOcc_2** | 0 | -11.4180 | 0.1171 | 8.7836 | 0.0431 |
| **LH_DorsAttnA_ParOcc_1** | 0 | -12.3090 | 0.1402 | 9.1811 | 0.0566 |
| **LH_DorsAttnA_SPL_1** | 0 | -10.7800 | 0.0647 | 8.5761 | 0.0383 |
| **LH_DorsAttnA_SPL_2** | 0 | -11.5950 | 0.1787 | 8.9260 | 0.0479 |
| **LH_DorsAttnA_SPL_3** | 0 | -10.2590 | 0.1365 | 8.6791 | 0.0488 |
| **LH_DorsAttnB_PostC_1** | 0 | -11.2640 | 0.1991 | 9.5755 | 0.0434 |
| **LH_DorsAttnB_PostC_2** | 0 | -10.7810 | 0.0252 | 9.2789 | 0.0616 |
| **LH_DorsAttnB_PostC_3** | 0 | -11.4290 | 0.0240 | 8.8249 | 0.0472 |
| **LH_DorsAttnB_PostC_4** | 0 | -11.5220 | 0.0843 | 9.2078 | 0.0570 |
| **LH_DorsAttnB_FEF_1** | 0 | -11.0750 | 0.1424 | 8.8283 | 0.0323 |
| **LH_SalVentAttnA_ParOper_1** | 0 | -11.6910 | 0.3255 | 9.3464 | 0.0631 |
| **LH_SalVentAttnA_Ins_1** | 0 | -8.3016 | 0.0077 | 6.9172 | -0.0141 |
| **LH_SalVentAttnA_FrOper_1** | 0 | -8.1766 | 0.1187 | 7.4587 | 0.0020 |
| **LH_SalVentAttnA_FrOper_2** | 0 | -8.5608 | 0.0932 | 7.8880 | -0.0023 |
| **LH_SalVentAttnA_ParMed_1** | 0 | -8.0986 | 0.0256 | 7.6081 | -0.0099 |
| **LH_SalVentAttnA_FrMed_1** | 0 | -7.4069 | -0.0637 | 6.6809 | -0.0060 |
| **LH_SalVentAttnA_FrMed_2** | 0 | -8.0962 | 0.0606 | 6.9157 | 0.0069 |
| **LH_SalVentAttnB_IPL_1** | 0 | -7.9181 | 0.1392 | 7.0358 | 0.0118 |
| **LH_SalVentAttnB_PFCl_1** | 0 | -8.4159 | 0.0540 | 6.9949 | -0.0147 |
| **LH_SalVentAttnB_Ins_1** | 0 | -8.5730 | -0.0175 | 7.2244 | 0.0063 |
| **LH_SalVentAttnB_PFCmp_1** | 0 | -7.2495 | 0.1335 | 6.7455 | -0.0077 |
| **LH_LimbicB_OFC_1** | 0 | -8.1628 | 0.0250 | 7.1411 | -0.0345 |
| **LH_LimbicB_OFC_2** | 0 | -7.8038 | 0.0715 | 7.0341 | 0.0029 |
| **LH_LimbicA_TempPole_1** | 0 | -7.9976 | 0.0413 | 7.7241 | 0.0059 |
| **LH_LimbicA_TempPole_2** | 0 | -8.8983 | 0.0053 | 7.9092 | -0.0250 |
| **LH_LimbicA_TempPole_3** | 0 | -7.1380 | 0.0813 | 6.4101 | -0.0086 |
| **LH_LimbicA_TempPole_4** | 0 | -8.1314 | 0.0715 | 6.9587 | -0.0074 |
| **LH_ContA_Temp_1** | 0 | -8.5749 | 0.0347 | 8.4590 | 0.0032 |
| **LH_ContA_IPS_1** | 0 | -8.6890 | 0.0897 | 7.8141 | 0.0083 |
| **LH_ContA_IPS_2** | 0 | -8.7044 | 0.1021 | 7.7895 | -0.0047 |
| **LH_ContA_IPS_3** | 0 | -7.8582 | 0.0688 | 7.3594 | -5.2245e-04 |
| **LH_ContA_PFCd_1** | 0 | -7.6873 | 0.0081 | 6.8786 | -0.0021 |
| **LH_ContA_PFClv_1** | 0 | -7.8211 | 0.0046 | 6.8185 | -0.0134 |
| **LH_ContA_PFCl_1** | 0 | -8.0065 | 0.0686 | 7.2162 | -0.0062 |
| **LH_ContA_PFCl_2** | 0 | -8.4473 | 0.0261 | 7.5265 | 0.0108 |
| **LH_ContA_PFCl_3** | 0 | -7.6784 | 0.0453 | 6.5892 | -0.0022 |
| **LH_ContA_Cingm_1** | 0 | -8.7581 | 0.1237 | 7.5396 | 0.0050 |
| **LH_ContB_Temp_1** | 0 | -7.6398 | 0.0605 | 6.7418 | -0.0011 |
| **LH_ContB_IPL_1** | 0 | -8.5475 | 0.1027 | 7.4685 | -0.0155 |
| **LH_ContB_PFCl_1** | 0 | -8.3619 | 0.1478 | 7.9688 | 0.0107 |
| **LH_ContB_PFClv_1** | 0 | -8.1948 | -0.0145 | 7.4539 | -4.3009e-04 |
| **LH_ContB_PFClv_2** | 0 | -8.2727 | 0.0665 | 7.7121 | 0.0025 |
| **LH_ContC_pCun_1** | 0 | -8.4333 | 0.0444 | 8.1789 | 0.0014 |
| **LH_ContC_pCun_2** | 0 | -8.0428 | 0.0084 | 7.0699 | 0.0081 |
| **LH_ContC_Cingp_1** | 0 | -8.1206 | 0.0637 | 7.4763 | 0.0046 |
| **LH_DefaultA_IPL_1** | 0 | -7.3273 | -0.0145 | 6.6384 | -6.9602e-04 |
| **LH_DefaultA_PFCd_1** | 0 | -8.1112 | -0.0123 | 7.2959 | 0.0063 |
| **LH_DefaultA_pCunPCC_1** | 0 | -7.6906 | 0.0478 | 7.3377 | 0.0062 |
| **LH_DefaultA_pCunPCC_2** | 0 | -7.7861 | 0.1348 | 7.0694 | -0.0026 |

| | | | | | |
|---|---|---|---|---|---|
| **LH_DefaultA_pCunPCC_3** | 0 | -8.1359 | -0.0026 | 7.0543 | 0.0088 |
| **LH_DefaultA_PFCm_1** | 0 | -8.7401 | 0.1343 | 7.7941 | -0.0034 |
| **LH_DefaultA_PFCm_2** | 0 | -6.7078 | -0.0428 | 6.6195 | -0.0025 |
| **LH_DefaultA_PFCm_3** | 0 | -7.0361 | -0.0747 | 6.9480 | 0.0039 |
| **LH_DefaultB_Temp_1** | 0 | -6.2994 | -0.0070 | 6.9565 | 0.0053 |
| **LH_DefaultB_Temp_2** | 0 | -6.1655 | -0.0422 | 6.4780 | -0.0097 |
| **LH_DefaultB_Temp_3** | 0 | -7.1990 | -0.0657 | 7.2563 | 0.0059 |
| **LH_DefaultB_Temp_4** | 0 | -7.7551 | -0.1953 | 7.1621 | 0.0126 |
| **LH_DefaultB_IPL_1** | 0 | -7.2994 | -0.1087 | 7.2810 | 0.0038 |
| **LH_DefaultB_PFCd_1** | 0 | -7.5027 | -0.1429 | 6.9655 | 0.0029 |
| **LH_DefaultB_PFCd_2** | 0 | -6.5027 | -0.0878 | 6.8308 | 0.0087 |
| **LH_DefaultB_PFCd_3** | 0 | -7.4422 | -0.0850 | 6.9591 | 0.0121 |
| **LH_DefaultB_PFCd_4** | 0 | -7.5244 | -0.1497 | 6.7982 | -0.0062 |
| **LH_DefaultB_PFCv_1** | 0 | -6.9080 | -0.0882 | 7.0308 | -0.0082 |
| **LH_DefaultB_PFCv_2** | 0 | -6.8427 | -0.1481 | 6.6485 | -0.0093 |
| **LH_DefaultB_PFCv_3** | 0 | -6.5632 | -0.1729 | 6.6345 | -0.0313 |
| **LH_DefaultB_PFCv_4** | 0 | -6.4749 | -0.0048 | 6.6313 | 0.0038 |
| **LH_DefaultC_IPL_1** | 0 | -6.9151 | -0.0164 | 6.6901 | 0.0157 |
| **LH_DefaultC_Rsp_1** | 0 | -5.7517 | 0.0027 | 5.8451 | 0.0132 |
| **LH_DefaultC_PHC_1** | 0 | -6.4839 | 0.0073 | 6.4530 | -0.0120 |
| **LH_TempPar_1** | 0 | -7.2836 | -0.0800 | 7.1065 | 0.0184 |
| **LH_TempPar_2** | 0 | -6.7864 | -0.0059 | 6.5391 | 0.0138 |
| **RH_VisCent_ExStr_1** | 0 | -7.1982 | -0.0792 | 6.8269 | 0.0205 |
| **RH_VisCent_ExStr_2** | 0 | -6.8952 | -0.1062 | 7.1474 | 0.0076 |
| **RH_VisCent_Striate_1** | 0 | -7.1558 | -0.1649 | 6.5753 | -0.0017 |
| **RH_VisCent_ExStr_3** | 0 | -6.6833 | -0.0820 | 6.8094 | -0.0015 |
| **RH_VisCent_ExStr_4** | 0 | -5.1460 | -0.0787 | 6.3008 | -0.0033 |
| **RH_VisCent_ExStr_5** | 0 | -6.7006 | -0.0017 | 6.7530 | -0.0233 |
| **RH_VisPeri_ExStrInf_1** | 0 | -6.7379 | -0.0085 | 6.8054 | 0.0059 |
| **RH_VisPeri_ExStrInf_2** | 0 | -6.5686 | -0.0887 | 6.9788 | -0.0088 |
| **RH_VisPeri_StriCal_1** | 0 | -7.5740 | -0.1354 | 7.2682 | -0.0042 |
| **RH_VisPeri_ExStrSup_1** | 0 | -6.2743 | -0.0713 | 6.7521 | -0.0055 |
| **RH_VisPeri_ExStrSup_2** | 0 | -6.1360 | -0.0061 | 6.2610 | 0.0048 |
| **RH_VisPeri_ExStrSup_3** | 0 | -6.8518 | -0.1326 | 6.9158 | 0.0019 |
| **RH_SomMotA_1** | 0 | -6.5202 | -0.0350 | 6.4978 | 0.0085 |
| **RH_SomMotA_2** | 0 | -7.3580 | 0.0420 | 7.3670 | 3.4309e-04 |
| **RH_SomMotA_3** | 0 | -7.4978 | 0.0095 | 7.1625 | 0.0027 |
| **RH_SomMotA_4** | 0 | -6.5866 | -0.1263 | 6.5705 | -0.0101 |
| **RH_SomMotA_5** | 0 | -5.7855 | 0.0356 | 6.1719 | -0.0191 |
| **RH_SomMotA_6** | 0 | -7.2777 | -0.1738 | 7.1359 | -4.9443e-04 |
| **RH_SomMotA_7** | 0 | -6.5366 | -0.0160 | 6.5622 | -0.0148 |
| **RH_SomMotA_8** | 0 | -6.2702 | 0.0597 | 6.3235 | -0.0144 |
| **RH_SomMotA_9** | 0 | -6.7530 | -0.0993 | 5.0383 | -0.0131 |
| **RH_SomMotA_10** | 0 | -5.5688 | -0.0480 | 4.3035 | -0.0165 |
| **RH_SomMotA_11** | 0 | -6.5020 | -0.1347 | 4.7690 | 0.0012 |
| **RH_SomMotB_Aud_1** | 0 | -5.0628 | 0.0095 | 4.2496 | -0.0052 |
| **RH_SomMotB_Aud_2** | 0 | -6.3936 | -0.0865 | 5.0733 | -0.0082 |
| **RH_SomMotB_S2_1** | 0 | -6.7003 | -0.0581 | 4.8917 | 0.0191 |
| **RH_SomMotB_S2_2** | 0 | -6.2726 | -0.1207 | 5.0602 | 0.0100 |
| **RH_SomMotB_S2_3** | 0 | -5.5381 | -0.0776 | 4.5322 | 0.0027 |
| **RH_SomMotB_S2_4** | 0 | -5.8264 | -0.0727 | 4.7658 | 0.0141 |
| **RH_SomMotB_Cent_1** | 0 | -6.7545 | -0.0133 | 4.8967 | 0.0064 |
| **RH_DorsAttnA_TempOcc_1** | 0 | -6.8440 | -0.0572 | 5.1954 | -0.0057 |
| **RH_DorsAttnA_ParOcc_1** | 0 | -5.9452 | -0.0010 | 4.9742 | 0.0195 |
| **RH_DorsAttnA_SPL_1** | 0 | -5.7175 | 0.0447 | 4.7394 | 0.0091 |
| **RH_DorsAttnA_SPL_2** | 0 | -7.0855 | 0.0125 | 4.8059 | 0.0160 |
| **RH_DorsAttnA_SPL_3** | 0 | -6.6885 | -0.0555 | 4.6890 | -0.0049 |
| **RH_DorsAttnA_SPL_4** | 0 | -7.2136 | -0.1302 | 5.1151 | 0.0017 |
| **RH_DorsAttnB_PostC_1** | 0 | -5.1889 | -0.0455 | 4.4941 | -0.0053 |
| **RH_DorsAttnB_PostC_2** | 0 | -5.7470 | -0.0020 | 4.5428 | 0.0050 |
| **RH_DorsAttnB_PostC_3** | 0 | -5.6985 | 0.0310 | 4.7601 | -1.1811e-04 |
| **RH_DorsAttnB_PostC_4** | 0 | -5.6875 | -0.0702 | 4.5617 | 0.0280 |
| **RH_DorsAttnB_FEF_1** | 0 | -6.4156 | -0.0445 | 4.8050 | 0.0281 |
| **RH_SalVentAttnA_ParOper_1** | 0 | -5.9861 | -0.0031 | 4.3800 | 0.0079 |

| | | | | | |
|---|---|---|---|---|---|
| **RH_SalVentAttnA_PrC_1** | 0 | -6.2884 | -0.0573 | 5.1019 | 0.0187 |
| **RH_SalVentAttnA_Ins_1** | 0 | -6.1059 | -0.1450 | 4.7077 | -0.0117 |
| **RH_SalVentAttnA_Ins_2** | 0 | -6.1346 | -0.0122 | 4.6938 | 0.0126 |
| **RH_SalVentAttnA_FrOper_1** | 0 | -6.2727 | 0.0052 | 5.1773 | 0.0283 |
| **RH_SalVentAttnA_FrMed_1** | 0 | -6.3836 | -0.0368 | 4.9134 | 0.0015 |
| **RH_SalVentAttnA_ParMed_1** | 0 | -5.7848 | -0.1242 | 5.0235 | 0.0284 |
| **RH_SalVentAttnA_ParMed_2** | 0 | -6.3048 | -0.0689 | 4.7339 | 0.0141 |
| **RH_SalVentAttnA_FrMed_2** | 0 | -5.9528 | -0.0419 | 5.0415 | 0.0011 |
| **RH_SalVentAttnB_IPL_1** | 0 | -3.2755 | -0.2115 | 3.9775 | 0.0055 |
| **RH_SalVentAttnB_PFClv_1** | 0 | -4.1598 | -0.0712 | 3.6288 | 0.0177 |
| **RH_SalVentAttnB_PFCl_1** | 0 | -4.4241 | -0.0825 | 4.9286 | 0.0262 |
| **RH_SalVentAttnB_Ins_1** | 0 | -4.0642 | -0.2091 | 4.2671 | 0.0148 |
| **RH_SalVentAttnB_Ins_2** | 0 | -4.5035 | -0.0847 | 4.6941 | 0.0011 |
| **RH_SalVentAttnB_PFCmp_1** | 0 | -3.7555 | -0.1653 | 3.8102 | 0.0221 |
| **RH_LimbicB_OFC_1** | 0 | -3.8251 | -0.0785 | 4.8340 | 0.0496 |
| **RH_LimbicB_OFC_2** | 0 | -3.6836 | 0.0241 | 3.3035 | 0.0278 |
| **RH_LimbicB_OFC_3** | 0 | -4.2895 | -0.1124 | 4.6011 | 0.0190 |
| **RH_LimbicB_OFC_4** | 0 | -3.7708 | 0.1173 | 3.4130 | 0.0052 |
| **RH_LimbicA_TempPole_1** | 0 | -3.6785 | -0.0705 | 3.8930 | 0.0321 |
| **RH_LimbicA_TempPole_2** | 0 | -3.6146 | 0.0498 | 3.3739 | 0.0217 |
| **RH_LimbicA_TempPole_3** | 0 | -3.0071 | 0.0974 | 3.3589 | 0.0154 |
| **RH_LimbicA_TempPole_4** | 0 | -4.4454 | -0.1904 | 5.1321 | 0.0144 |
| **RH_ContA_IPS_1** | 0 | -3.7697 | -0.2095 | 3.8292 | 0.0215 |
| **RH_ContA_IPS_2** | 0 | -3.2596 | -0.0720 | 3.5382 | 0.0246 |
| **RH_ContA_PFCd_1** | 0 | -3.2797 | 0.1803 | 3.9886 | 0.0140 |
| **RH_ContA_PFCl_1** | 0 | -4.5347 | -0.2172 | 4.8364 | 0.0220 |
| **RH_ContA_PFCl_2** | 0 | -3.6422 | 0.1035 | 3.2344 | 0.0036 |
| **RH_ContA_Cingm_1** | 0 | -3.0874 | 0.1278 | 2.7804 | 0.0132 |
| **RH_ContB_Temp_1** | 0 | -3.4631 | 0.2083 | 2.8291 | 0.0052 |
| **RH_ContB_Temp_2** | 0 | -3.7244 | -0.0974 | 3.7493 | 0.0136 |
| **RH_ContB_IPL_1** | 0 | -3.8615 | 0.0602 | 3.6271 | 0.0138 |
| **RH_ContB_IPL_2** | 0 | -4.2661 | 0.0285 | 4.5688 | 0.0111 |
| **RH_ContB_PFCld_1** | 0 | -3.9685 | 0.1950 | 3.4609 | 0.0093 |
| **RH_ContB_PFCld_2** | 0 | -9.7144 | 0.0934 | 9.0460 | -0.0061 |
| **RH_ContB_PFClv_1** | 0 | -8.6469 | 0.0818 | 7.9645 | 0.0310 |
| **RH_ContB_PFClv_2** | 0 | -9.0572 | 0.2004 | 8.3942 | 0.0174 |
| **RH_ContB_PFCmp_1** | 0 | -9.1861 | 0.0674 | 8.4406 | 0.0127 |
| **RH_ContB_PFCld_3** | 0 | -9.1254 | -0.0148 | 8.7876 | 0.0016 |
| **RH_ContC_pCun_1** | 0 | -8.8513 | -0.0591 | 8.4487 | 0.0022 |
| **RH_ContC_pCun_2** | 0 | -8.7174 | 0.0466 | 8.4205 | -0.0092 |
| **RH_ContC_Cingp_1** | 0 | -9.6645 | 0.0469 | 8.7802 | 0.0131 |
| **RH_DefaultA_IPL_1** | 0 | -8.7193 | 0.0161 | 8.4664 | 0.0135 |
| **RH_DefaultA_PFCd_1** | 0 | -9.4870 | -0.0046 | 8.8954 | 0.0046 |
| **RH_DefaultA_pCunPCC_1** | 0 | -9.0957 | 0.0027 | 8.5463 | 0.0113 |
| **RH_DefaultA_PFCm_1** | 0 | -9.4930 | 0.0327 | 8.6518 | 0.0117 |
| **RH_DefaultA_PFCm_2** | 0 | -9.0158 | 0.0207 | 8.0444 | 0.0240 |
| **RH_DefaultA_PFCm_3** | 0 | -9.4765 | 0.0389 | 8.7927 | 0.0282 |
| **RH_DefaultB_Temp_1** | 0 | -7.7183 | 0.0534 | 7.3822 | 0.0178 |
| **RH_DefaultB_AntTemp_1** | 0 | -9.3912 | -0.0177 | 8.4604 | 0.0146 |
| **RH_DefaultB_PFCd_1** | 0 | -9.3421 | 0.1495 | 8.4465 | 0.0282 |
| **RH_DefaultB_PFCv_1** | 0 | -10.2300 | -0.0142 | 9.3770 | 0.0091 |
| **RH_DefaultC_IPL_1** | 0 | -9.0098 | 0.2499 | 8.6093 | 0.0149 |
| **RH_DefaultC_Rsp_1** | 0 | -9.0898 | 0.0517 | 8.6129 | 0.0227 |
| **RH_DefaultC_PHC_1** | 0 | -8.7774 | -0.0752 | 8.2326 | 0.0300 |
| **RH_TempPar_1** | 0 | -9.3009 | 0.0599 | 8.6732 | 0.0093 |
| **RH_TempPar_2** | 0 | -8.8858 | 0.0909 | 8.1276 | -0.0014 |
| **RH_TempPar_3** | 0 | -9.3544 | 0.0521 | 8.4929 | 0.0173 |
| **RH_TempPar_4** | 0 | -9.2239 | 0.0988 | 8.5997 | 0.0210 |

**Questions:**

➡ 1. How many columns are there?

- ANS: 201 columns

➡ 2. What do the columns mean?

- ANS: column 1 represents sampling time points, and column 2-201 represents MEG activity from the 200 brain regions

➡ 3. How many rows?

- ANS: 11150 rows

➡ 4. What do the rows mean?

- ANS: the rows represent a time sample in the MEG recording

**Extract values into matrices**

```
%% Code Examples
A = tbl.ColName;    % let A be a N x 1 matrix consisting of the values from the column ColName
of table tbl

A = table2array(tbl);   % let A be a N x M matrix consisting of all numerical values from the
N rows and M columns of table tbl

B = A(5:10, 2:6);       % let B be a 6x5 matrix containing the numbers from the 5th to 10th
rows and the 2nd to 6th columns of matrix A
B = A(1:end, 2:6);      % let B be an Nx5 matrix containing the numbers from all rows and the
2nd to 6th columns of matrix A
```

➡ **Extract** the time into a N x 1 matrix named `time`

```
time = meg_table{:,1};
```

➡ **Extract** the recorded signals into a matrix named S

```
S = meg_table{:,2:end};
```

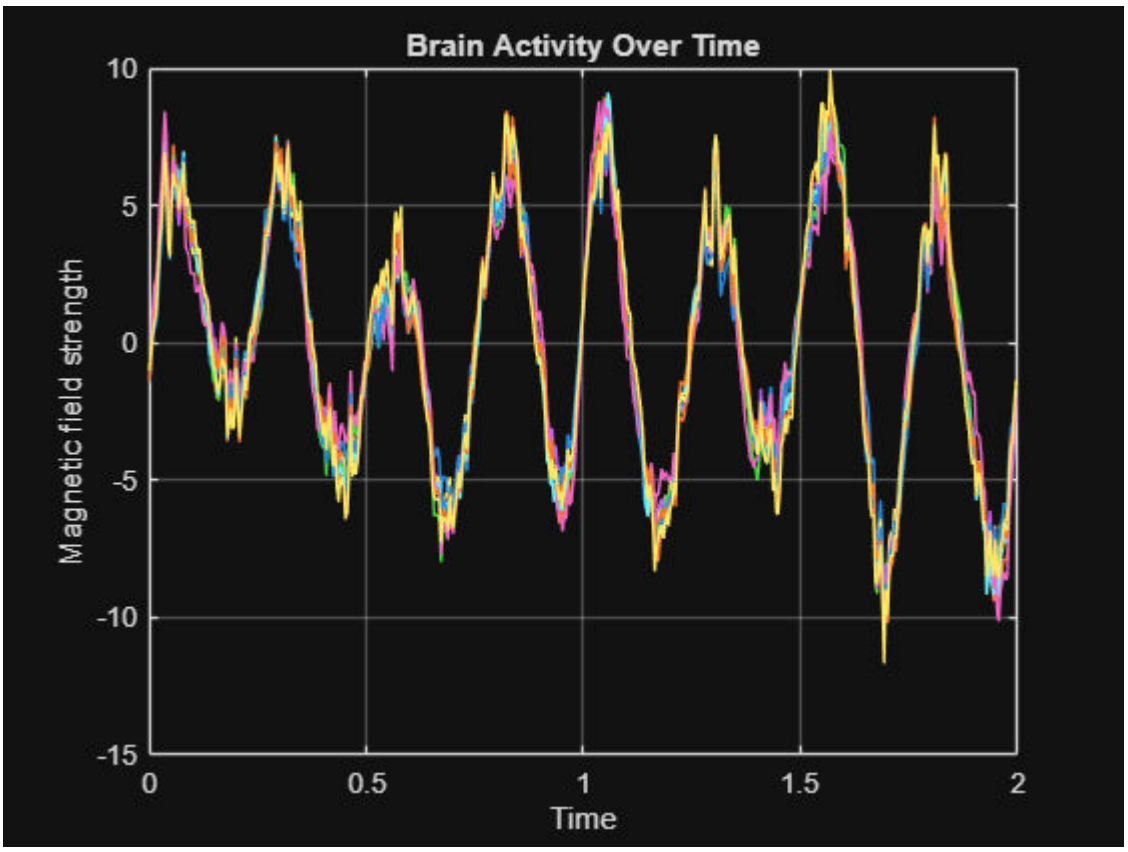
**Visualize the time series data**

Let's take a look at the data.

➡ Use MATLAB's plotting tools to generate a **line plot:**

- showing the brain activity over time
- with magnetic field strength on the y-axis, and
- time on the x-axis
- make sure to label your axes!

```
figure;
plot(time, S(:,1:10))
xlabel('Time')
ylabel('Magnetic field strength')
title('Brain Activity Over Time')
grid on;
```




Although we can qualitatively observe a pattern in the data, notice how when $R$ is large, visualizing or comparing all regions at once becomes challenging.

## Problem 2 - Transformations: Aligning Brains into a Common Space

Every person's head is positioned slightly differently inside the MEG helmet. A subject's head may be shifted, rotated, or scaled compared to a standard template. To compare brain data across individuals, neuroscientists must place all brains into the same coordinate system.

We do this using a 3×3 transformation matrix, which performs a combination of rotation, scaling, shear, and reflection. In this problem, you will use anatomical landmarks, the nose and two ear points, to compute the transformation that maps a subject's head position into template space, and then apply it to all of the subject's brain coordinates. This step mirrors what neuroscientists do before any analysis can begin.

### Compute the Transformation Matrix

Suppose we know three corresponding landmark points:

- Nose
- Left ear
- Right ear

For each point, we have a **source coordinate** (from the subject) and a **target coordinate** (from the template).

We want to find a transformation matrix $A$ such that:

$$A\, p_{\text{source}} = p_{\text{target}}$$

### Given the following coordinates

```
% Source and target coordinates
target_nose_mm = [1; 91; -28];
source_nose_mm = [1.5; 85.3; -20.2];

target_earl_mm = [-83; -19; -47];
source_earl_mm = [-81.7; -15.6; -53.0];

target_earr_mm = [83; -19; -47];
source_earr_mm = [85.4; -16.2; -50.7];
```

➡ **Compute the transformation matrix** $A$ that will transform the source landmark points to the target landmark points.

```
p_source = [source_nose_mm, source_earl_mm, source_earr_mm];
p_target = [target_nose_mm, target_earl_mm, target_earr_mm];

A = p_target / p_source;
```

### Apply the Transformation Matrix

➡ Use the matrix you computed to **transform the brain coordinates** you loaded in Problem 1 from source subject space into the target template space.

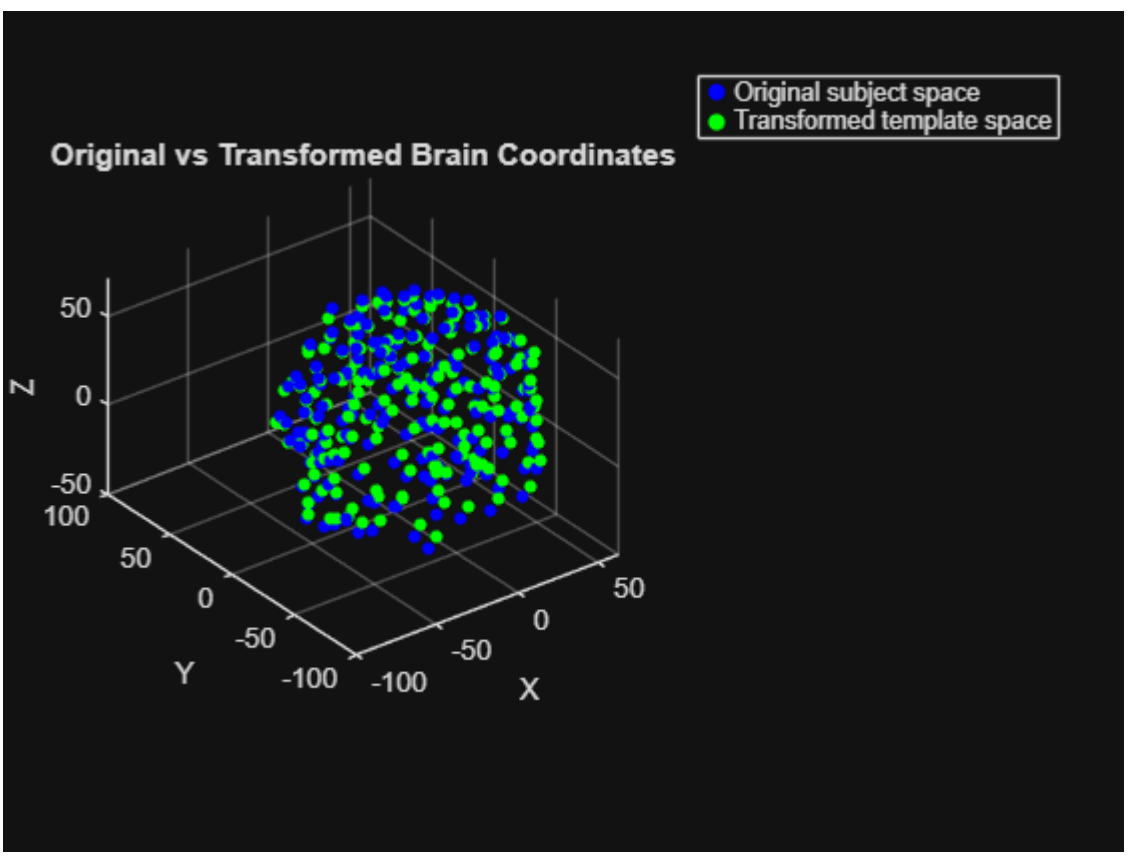
```
L_transformed = (A * L')';
```

**Plot the Original and Transformed Brains Together**

➡ In a new figure, plot both point clouds together on the same plot as 3D scatter plots.

- ensure the subject-space and template-space point clouds are shown in different colours
- ensure these colours are labelled in a legend

```
figure;
scatter3(L(:,1), L(:,2), L(:,3), 20, 'b', 'filled');
hold on;
scatter3(L_transformed(:,1), L_transformed(:,2), L_transformed(:,3), 20,
'g', 'filled');
xlabel('X'); ylabel('Y'); zlabel('Z');
title('Original vs Transformed Brain Coordinates');
legend('Original subject space', 'Transformed template space');
grid on;
daspect([1 1 1]);
```



## Problem 3 - Dimensionality reduction using Principle Component Analysis (PCA)

When recording electrical activity from multiple brain regions simultaneously (e.g., 200 recording sites), direct visualization of all signals becomes impractical. However, inspection of such data often reveals structure: subsets of brain regions exhibit correlated activity patterns, suggesting coordinated neural processes rather than independent signals.

### Objective

Principal Component Analysis (PCA) provides a systematic approach to identify and extract these dominant patterns of covariation. Rather than analyzing 200 individual timeseries, PCA allows us to represent the data using a smaller number of components (e.g., 5-10) where each component describes:

1. A temporal pattern of activity (component timeseries)
2. A spatial pattern indicating which brain regions participate in this activity and their relative contributions (component loadings)

### Mathematical Framework

### Quantifying Covariation

The covariance between two signals quantifies their degree of linear association. For signals *i* and *j*, the covariance measures how their fluctuations align over time. We compute covariance for all pairs of recording sites, yielding a **covariance matrix C** where each element $C_{ij}$ represents the normalized pairwise relationship between signals *i* and *j*.

### Eigendecomposition

The eigendecomposition of the covariance matrix identifies its intrinsic structure:

- **Eigenvectors** define orthogonal directions in the data space
- **Eigenvalues** quantify the variance explained along each corresponding eigenvector direction

Geometrically, the eigendecomposition reveals the principal axes of variation in the data. For a covariance matrix, eigenvectors indicate directions of maximal variance, while eigenvalues measure the magnitude of variance along those directions.

### Putting this all together

1. Neural signals exhibit covariation across recording sites
2. The covariance matrix quantifies pairwise relationships between all signals
3. Eigendecomposition of the covariance matrix identifies the dominant directions of variation, revealing which sets of brain regions covary and the relative magnitude of each pattern

The principal components, ordered by decreasing eigenvalue, provide a ranked decomposition of the data into orthogonal patterns. Typically, the first few components capture the majority of variance, enabling dimensionality reduction while preserving the essential structure of the neural activity.

Here we compute the covariance matrix for you:

```
% normalize each column
S_norm = zscore(S, [], 1);

% calculate the covariance matrix
C = cov(S_norm);
```

**Eigenvalue decomposition**

Next, we must compute the eigenvalue decomposition of the covariance matrix

$$C = V \Lambda V^T$$

where

- columns of $V$ are eigenvectors (principal directions),
- diagonal entries of $\Lambda$ are eigenvalues (variance explained).

The individual values in $V$ give the relative contribution of each brain region to the component.

- The rows of $V$ correspond to brain regions
- The columns of $V$ correspond to the principal components

In other words, the relative contribution (weight) of region $n$ to component $m$ is given by element $V_{n,m}$ in the eigenvector matrix.

➡ Use the relevant MATLAB function to:

1. **compute the Eigen decomposition**
2. **extract the Eigen values** as an N x 1 matrix

*Hint*: see code examples in Exercise 2

```
[V, D] = eig(C);
eigenvalues = diag(D);
```

➡ **Sort the eigenvalues** in descending order so that the components that explain the most variance (eg. the *principal* components) are first.

Store the sorted eigenvalues in a variable named `eigenvalues_sorted`

```
[eigenvalues_sorted, my_order] = sort(eigenvalues, 'descend');
```

➡ Use the sort index derived from sorting the eigenvalues to also **sort the matrix of eigenvectors** and store it into a variable named `eigenvectors_sorted`

```
eigenvectors_sorted = V(:, my_order);
```

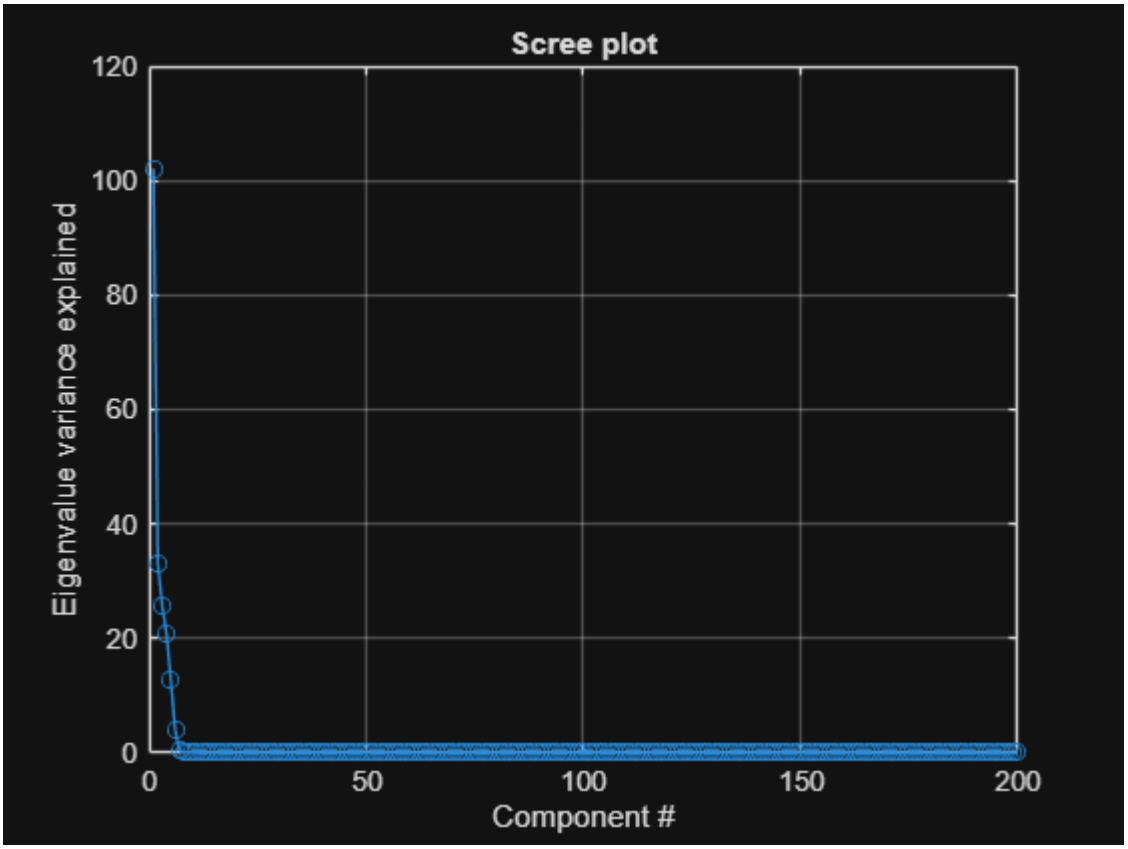
**Scree plot**

Plotting the eigenvalues gives us an idea of how much of the variance in our data is represented by the first few principal components.

- This plot is also a good sense check to make sure we computed everything correctly
- This plot is known as a "scree plot"

➡ In a new figure, **plot your sorted eigenvalues**.

```
figure;
plot(eigenvalues_sorted, '-o');
xlabel('Component #');
ylabel('Eigenvalue variance explained')
title('Scree plot')
grid on;
```




**Questions**

➡ 1. Based on your scree plot, qualitatively, how many components explain the majority of variance in your data?

ANS: There is a steep drop in eigenvalues after the first few components (~5) so around the first 5 components explain the majority of variance

➡ 2. Why might we want to keep only the first few components rather than all of them?

ANS: By only keeping the first few components, only the main patterns of neural activity are considered and noise/variations are minimized, making the analysis much more simple

**A visual representation of similarity in activity between brain regions**

The scree plot shows **how many components are important**.

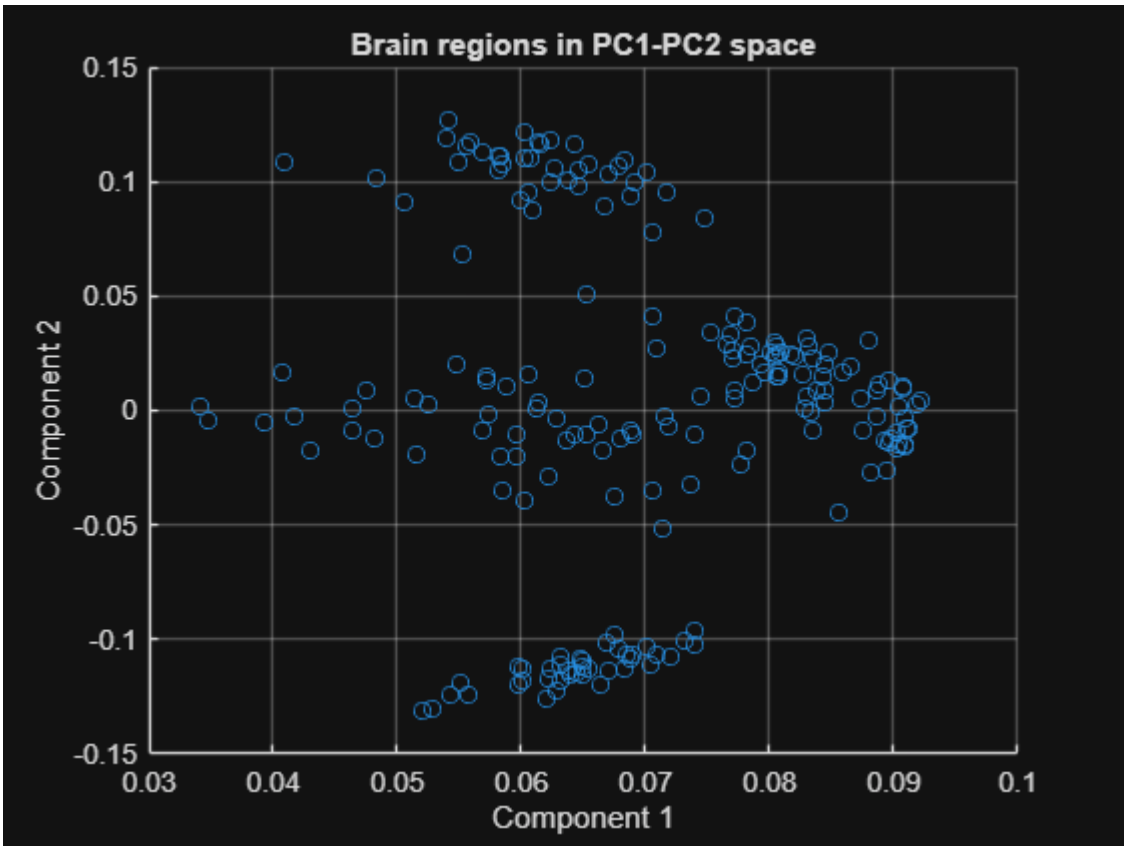
Now we want to answer a different question: **What do the important components actually look like?**

The eigenvectors tell us *how strongly each brain region reflects/participates in/contributes to each component*.

Here we plot the eigenvectors from the two components with highest variance.

```
figure;
scatter(eigenvectors_sorted(:,1), eigenvectors_sorted(:,2));
```

```
xlabel('Component 1');
ylabel('Component 2');
title('Brain regions in PC1-PC2 space');
grid on;
```




**Questions:**

➡ 1. In the PC1–PC2 loading plot, qualitatively, how many clusters of regions do you see?

ANS: I see around 3 clusters with similar patterns

➡ 2. What does it mean for two regions to be close in this PCA space?

ANS: If 2 regions are close in the PCA space their activity should be highly correlated, fluctuating together over time

**Plot a principal component's timeseries and relative brain region contribution**

Finally, let's visualize the timeseries and the brain regions involved in the first principal component.

➡ **Project** the sensor measurements into the principal component eigenspace using your sorted eigenvector matrix.

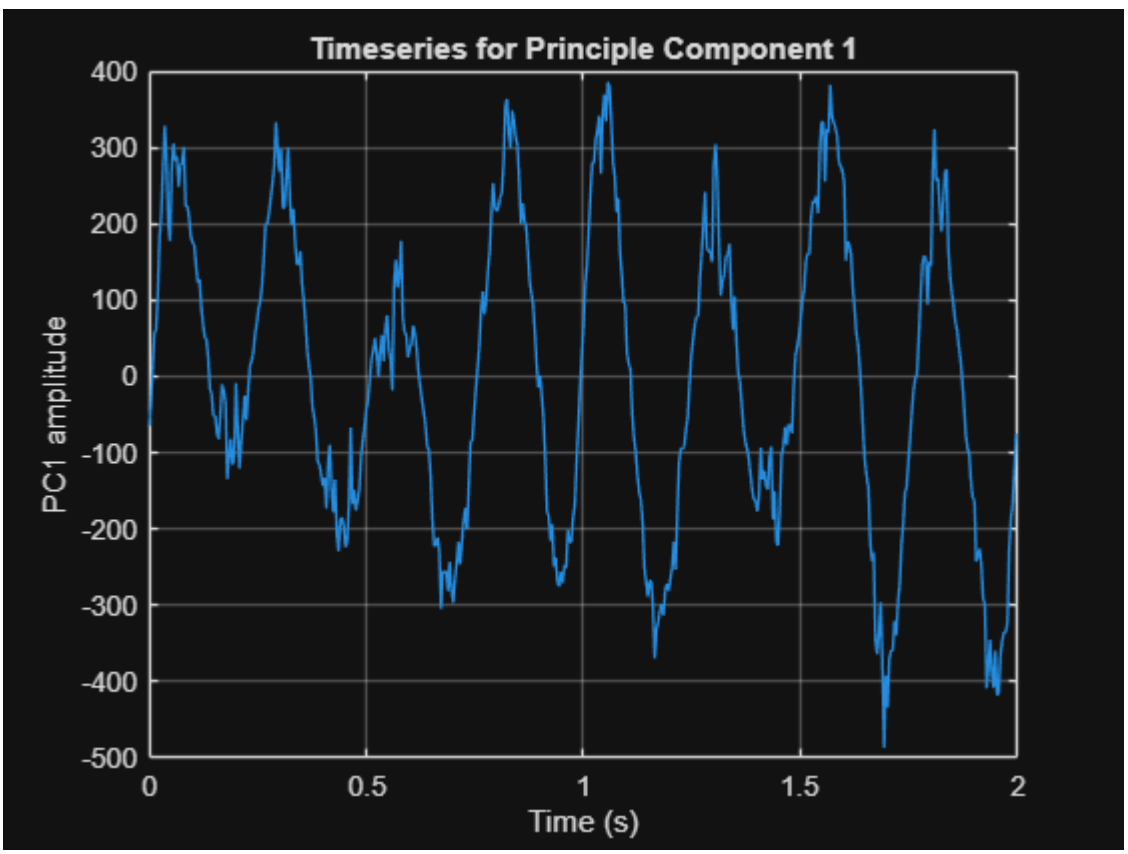
```
PC_scores = S_norm * eigenvectors_sorted;
```

➡ Extract the timeseries of the first principal component into a variable named `PC1_amplitude`

```
PC1_amplitude = PC_scores(:,1);
```

➡ Plot the timeseries over time with a line plot

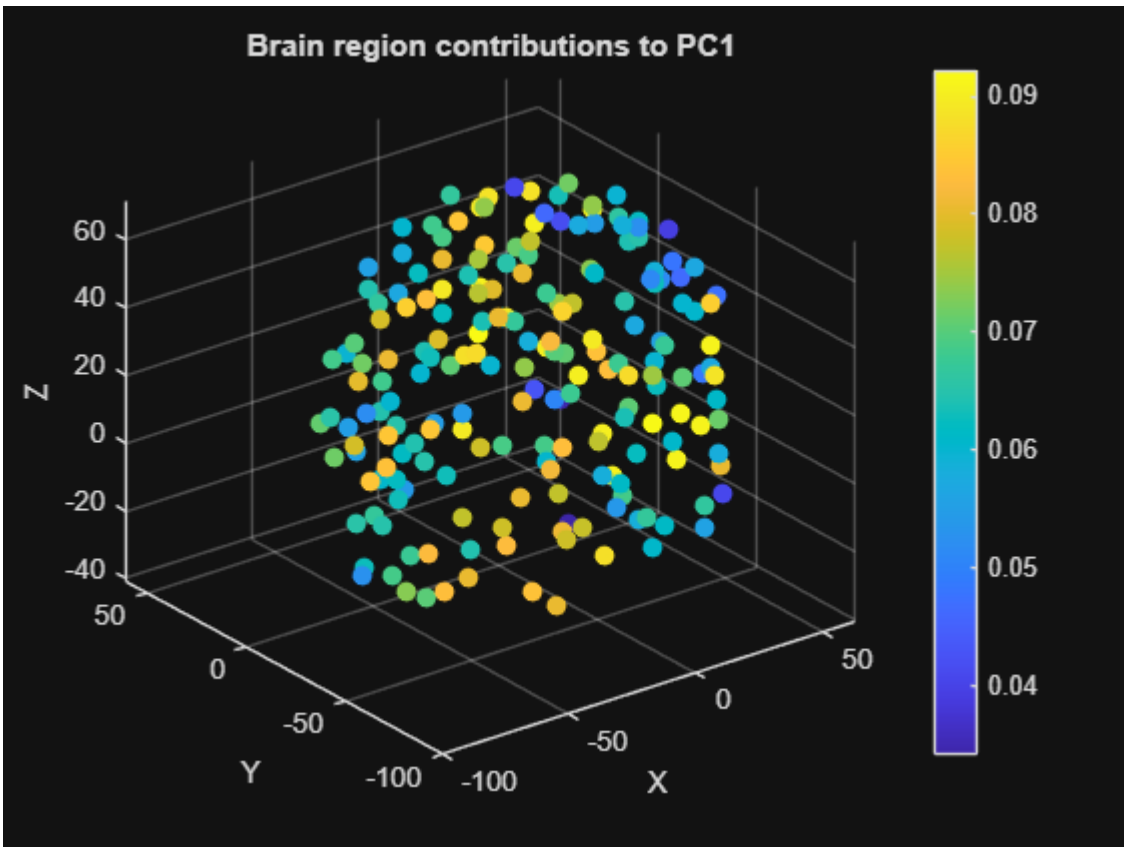
```
figure;
plot(time, PC1_amplitude);
xlabel('Time (s)');
ylabel('PC1 amplitude');
title('Timeseries for Principle Component 1')
grid on;
```




➡ Plot which brain regions contribute to this principal component

- Create a 3D scatter plot as above, using the template locations for brain regions
- Use the colours of each dot to represent the weight of that brain region within principal component 1

```
figure;
scatter3(L(:,1), L(:,2), L(:,3), 50, eigenvectors_sorted(:,1), 'filled');
xlabel('X');
ylabel('Y');
zlabel('Z');
title('Brain region contributions to PC1');
colorbar;
grid on;
```




## Problem 4 - Relating brain activity to measures of neurocognition

Suppose we imaged the brain activity of 50 healthy volunteers and separately asked them to complete a set of neurocognitive assessments that measure attention. Let's see whether there's a correlation between average brain activity in our component of interest and varying measures of domains of attention.

**Recall that:**

- **The least-squares problem** a mathematical technique used to find the best-fitting line or curve for a set of data points by minimizing the sum of the squares of the vertical distances (residuals) between each data point and the proposed model line.

- Specifically, the **least-squares problem** is finding $\vec{x} = \underset{x}{\operatorname{argmin}} \left\| A\vec{x} - b \right\|$ for an overdetermined system $A\vec{x} = \vec{b}$.
- The corresponding **normal equation,** $A^T A\vec{x} = A^T\vec{x}$, yields the least-squares (best-fit) solution, $\vec{x}^* = (A^T A)^{-1} A^T \vec{b}$, when $A^T A$ is invertible.

➡ First, load the pre-processed MEG data and assessment scores as a table from `https://github.com/dtxe/mat188_datasets/raw/refs/heads/main/meg/meg_with_neuropsych.csv` into a table named `meg_x_neuropsych`

```
meg_x_neuropsych = readtable('https://github.com/dtxe/mat188_datasets/raw/refs/heads/main/meg/meg_with_neuropsych.csv')
```

meg_x_neuropsych = 50×5 table

| | Age | TOVA_Score | PVT_Mean_RT_ms | ASRS_Score | MEG_PC1_fT |
|---|---|---|---|---|---|
| 1 | 12.2436 | 111.0770 | 304.8539 | 32.7295 | 310.4326 |
| 2 | 20.3100 | 102.5705 | 348.4322 | 45.6857 | 394.6329 |
| 3 | 17.2479 | 98.2653 | 264.8973 | 10.7102 | 339.8334 |
| 4 | 15.3812 | 95.4834 | 283.6169 | 32.2156 | 284.6462 |
| 5 | 9.1843 | 77.8222 | 280.3946 | 33.1186 | 204.5067 |
| 6 | 9.1839 | 89.2023 | 226.8243 | 39.3819 | 238.7769 |
| 7 | 7.8132 | 93.0904 | 314.8060 | 15.1566 | 209.9587 |
| 8 | 19.1265 | 115.8568 | 313.0528 | 14.1545 | 403.8159 |
| 9 | 15.4156 | 105.1543 | 300.2557 | 36.2633 | 336.5888 |
| 10 | 16.9130 | 73.5544 | 288.2706 | 33.5638 | 244.1374 |
| 11 | 7.2882 | 104.8613 | 229.2315 | 33.0059 | 260.6083 |
| 12 | 20.5787 | 94.2238 | 278.9677 | 34.1574 | 369.3795 |
| 13 | 18.6542 | 89.8462 | 282.8643 | 21.8397 | 349.0454 |
| 14 | 9.9727 | 109.1751 | 259.8861 | 32.7870 | 312.7331 |

⋮

```
summary(meg_x_neuropsych)
```

```
meg_x_neuropsych: 50×5 table

Variables:

    Age: double
    TOVA_Score: double
    PVT_Mean_RT_ms: double
    ASRS_Score: double
    MEG_PC1_fT: double

Statistics for applicable variables:

                      NumMissing        Min            Median           Max             Mean            St

    Age                    0            7.2882         13.1047         20.5787         13.2429         4.0
    TOVA_Score             0           60.7038        100.3842        123.4697         99.9558        13.7
    PVT_Mean_RT_ms         0          204.0614        297.2711        423.1621        297.1287        48.3
    ASRS_Score             0           10.7102         33.0623         62.6420         31.3171        11.2
    MEG_PC1_fT             0          204.5067        301.4661        416.4313        302.2798        55.5
```

```
disp(meg_x_neuropsych.Properties.VariableNames)
```

```
    {'Age'}    {'TOVA_Score'}    {'PVT_Mean_RT_ms'}    {'ASRS_Score'}    {'MEG_PC1_fT'}
```

### Compute the line of best fit between brain activity and age

➡ Using the `meg_x_neuropsych` table we loaded above, create $A$ and $b$.

```
age = meg_x_neuropsych.Age;
pc1_activity = meg_x_neuropsych.MEG_PC1_fT;

b = pc1_activity;
A = [age, ones(size(age))];
```

➡ **Compute the linear equation** that best represents the relationship between age as the independent variable (x) and brain activity in principal component 1 (PC1) as the dependent variable (y) using the matrix equation for the least-squares problem.

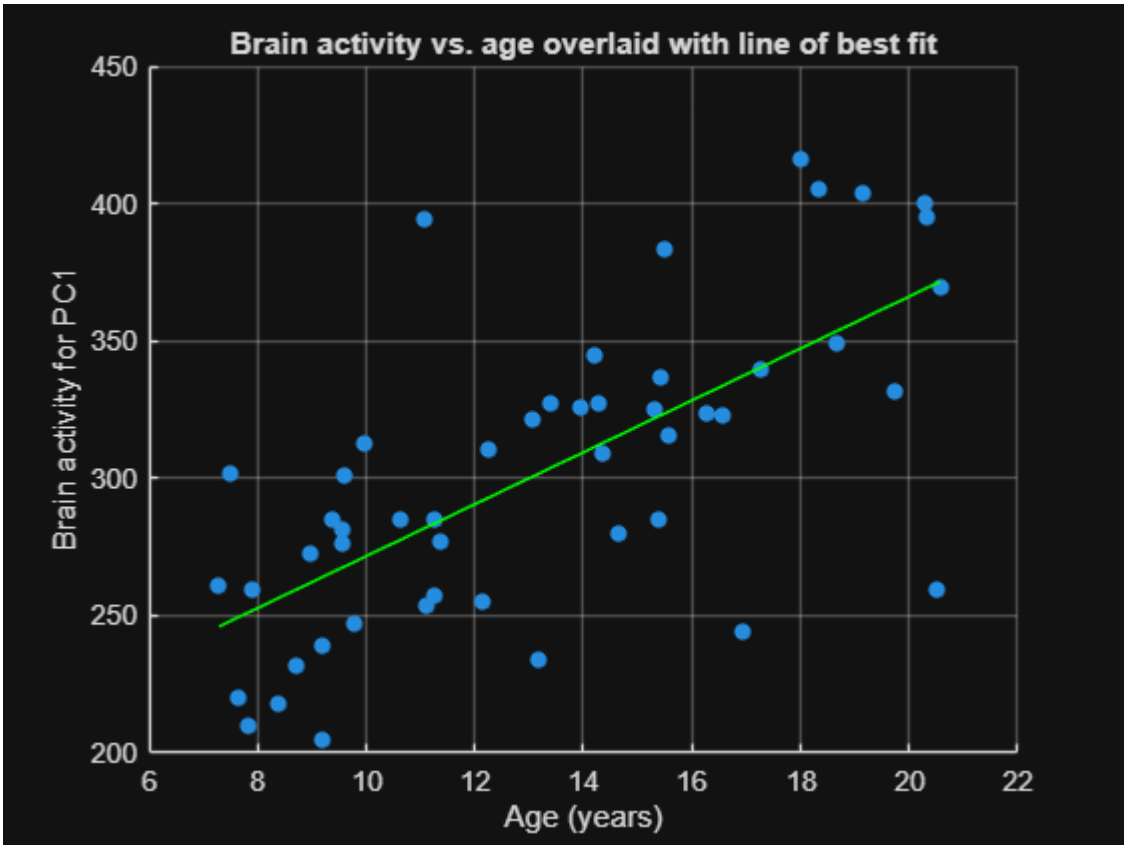
```
age_coeff = (A' * A) \ (A' * b);
```

### Generate a scatter plot

➡ In a new figure, **plot**:

- a scatter plot of brain activity vs. age
- overlaid with the line of best fit

```
figure;
scatter(age, pc1_activity, 'filled');
hold on;
plot(age, A*age_coeff, 'g');
xlabel('Age (years)');
ylabel('Brain activity for PC1');
title('Brain activity vs. age overlaid with line of best fit');
grid on;
hold off;
```




**Compute the line of best fit between brain activity and neurocognitive assessment scores**

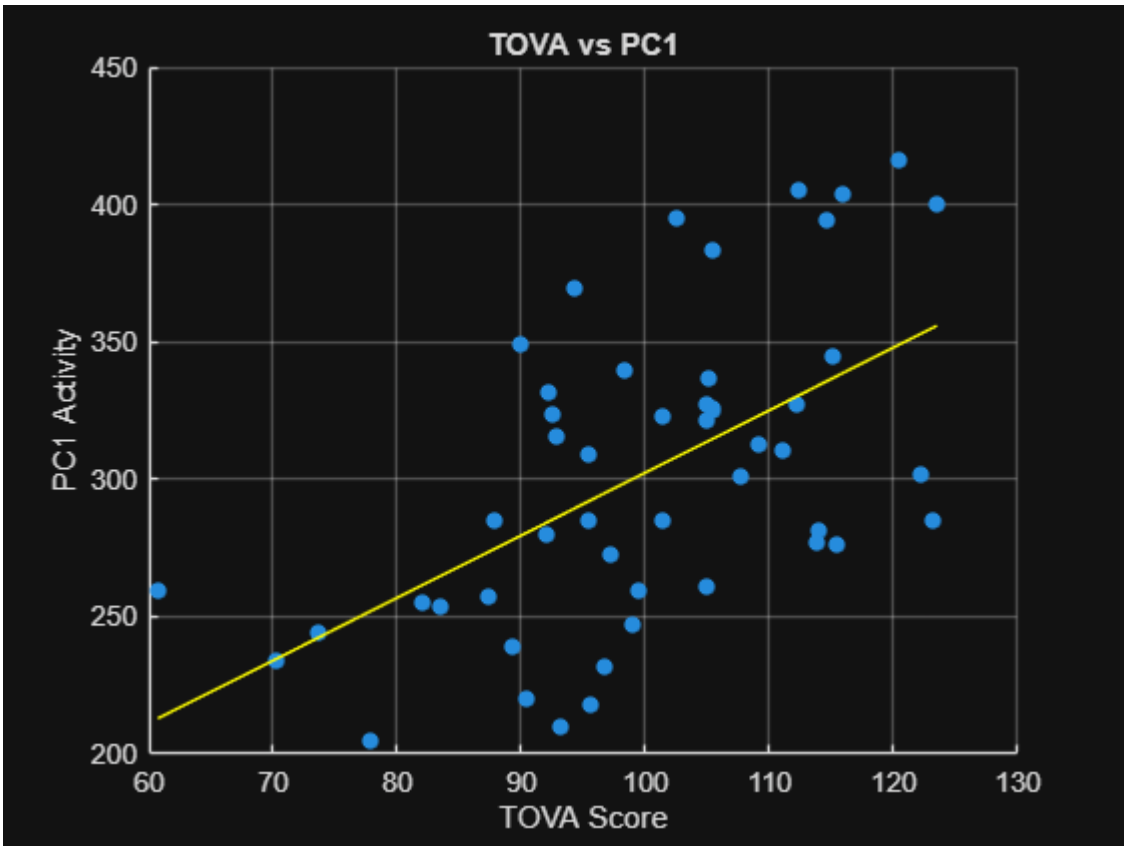
➡ Repeat the line of best fit individually / pair-wise between brain activity and the 3 neurocognitive assessments provided.

```
tova = meg_x_neuropsych.TOVA_Score;
pvt = meg_x_neuropsych.PVT_Mean_RT_ms;
asrs = meg_x_neuropsych.ASRS_Score;

A_tova = [tova, ones(size(tova))];
tova_fit = (A_tova' * A_tova) \ (A_tova' * pc1_activity);

A_pvt = [pvt, ones(size(pvt))];
pvt_fit = (A_pvt' * A_pvt) \ (A_pvt' * pc1_activity);
```

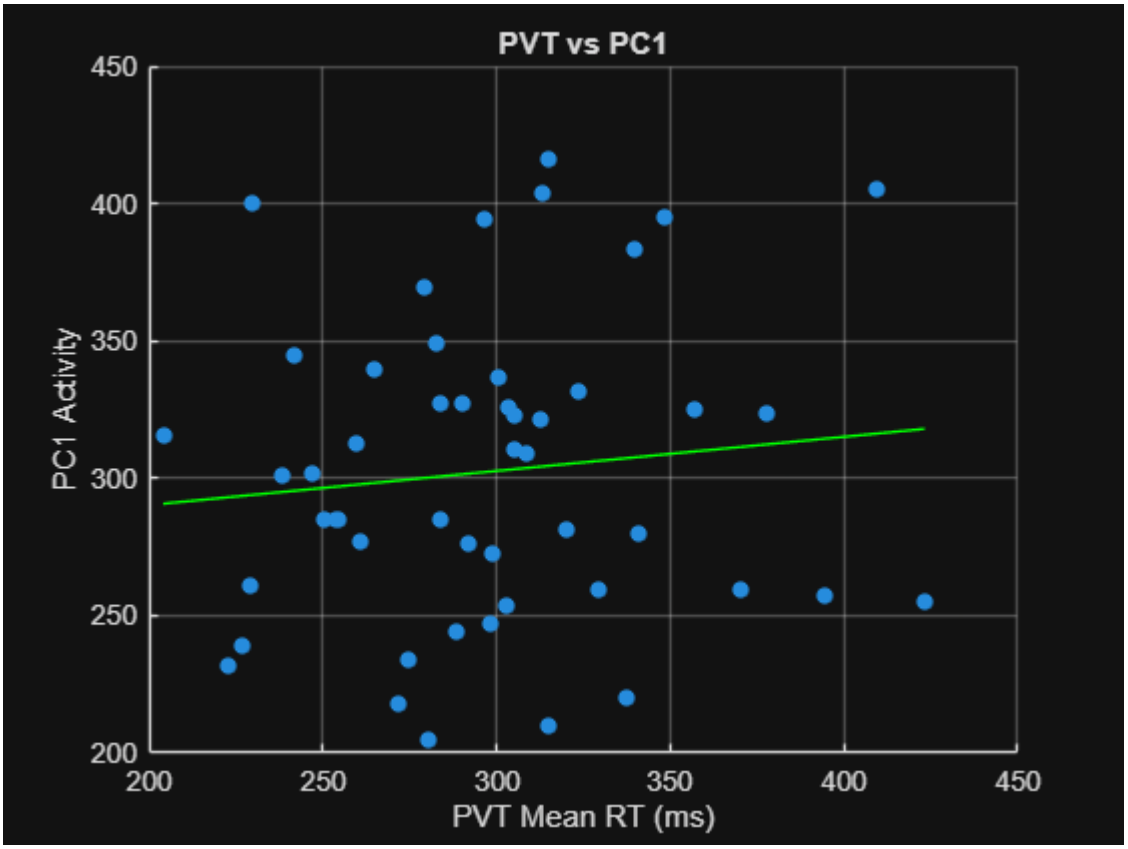
```
A_asrs = [asrs, ones(size(asrs))];
asrs_fit = (A_asrs' * A_asrs) \ (A_asrs' * pc1_activity);
```

➡ Plot lines of best fit on top of scatter plots for all variables

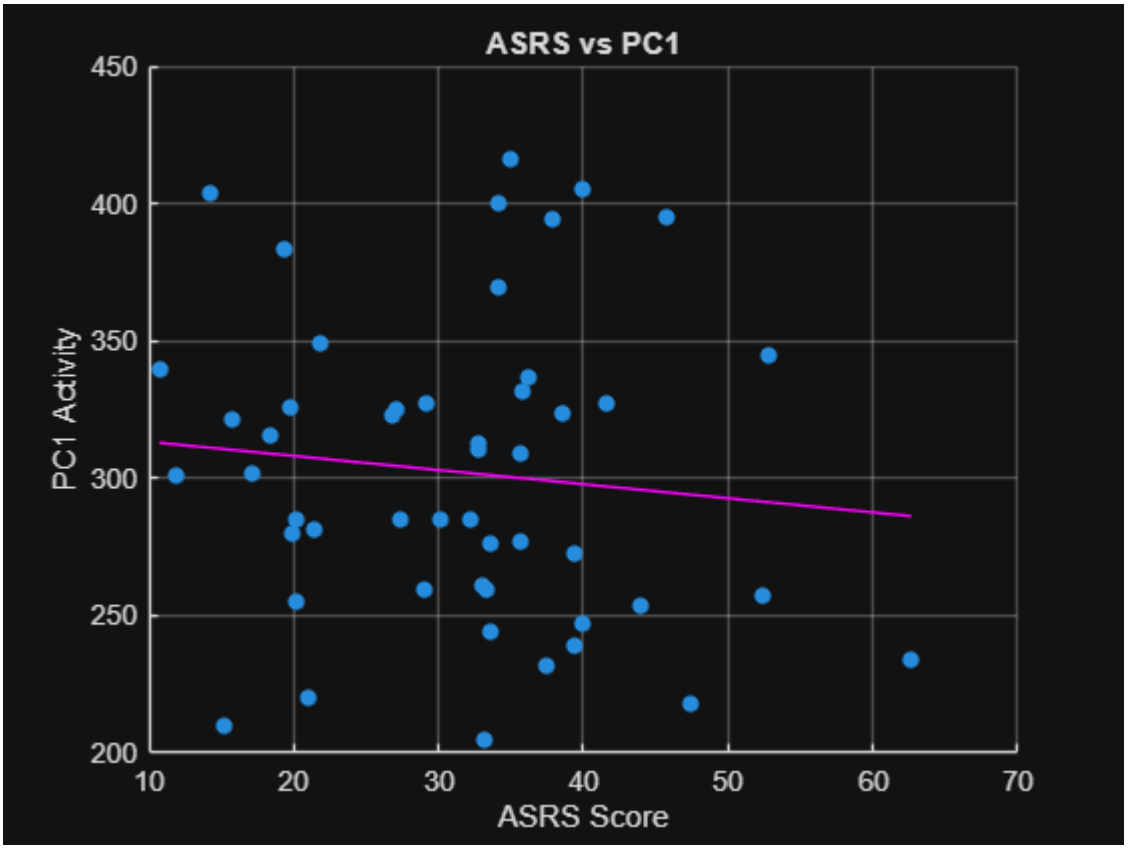
```
figure;
scatter(tova, pc1_activity, 'filled');
hold on;
plot(tova, tova_fit(1)*tova + tova_fit(2), 'y-');
xlabel('TOVA Score');
ylabel('PC1 Activity');
title('TOVA vs PC1');
grid on;
```




```
figure;
scatter(pvt, pc1_activity, 'filled');
hold on;
plot(pvt, pvt_fit(1)*pvt + pvt_fit(2), 'g-');
xlabel('PVT Mean RT (ms)');
ylabel('PC1 Activity');
title('PVT vs PC1');
grid on;
```




```
figure;
scatter(asrs, pc1_activity, 'filled');
hold on;
plot(asrs, asrs_fit(1)*asrs + asrs_fit(2), 'm-');
xlabel('ASRS Score');
ylabel('PC1 Activity');
title('ASRS vs PC1');
grid on;
```




**Interpretation**

➡ 1. Which variable(s) correlate with brain activity, qualitatively?

ANS: TOVA evidently has a positive correlation with brain activity as the line has a strong positive slope. This means that higher TOVA scores are associated with higher PC1 activity, and vice versa. PVT mean RT has a little bit of correlation, although not that much as the slope is only slightly positive.

➡ 2. Which variable(s) do not correlate with brain activity, qualitatively?

ANS: ASRS does not correlate with brain activity as it has a nearly horizontal line with slight negative slope. Additionally, the scatter points are all over the place and don't trend towards a certain area, so no valid correlation can be deduced qualitatively.

➡ 3. Based on your results, what can we say, qualitatively, about brain activity in PC1 and attentive function in adults? What about in adolescents?

ANS: We can first note that the main correlation we found with PC1 was with TOVA. Furthermore, PVT had a weaker correlation. This suggests that TOVA scores may be more indicative of PC1 scores. Additionally, as there is no correlation with ASRS, we can ultimately focus on the effect of TOVA for adults and adolescents. As TOVA relates to attentive function, it is reasonable to say that adults would probably develop more attentive

skills compared to adolescents. As such, we can say qualitatively that adults would have stronger PC1 activity compared to adolescents, who have not fully developed their attentive function.

# Congratulations!

**You've successfully completed the MATLAB Assignment! ⁇**

We hope this provided a demonstration of the power of linear algebra and computation software in working with and manipulating, isolating trends in, and extracting insights from large datasets.

## Submission instructions

1. Run this Live Script from top to bottom to verify the correctness of your code. (Home tab > Clear Workspace; Live Editor tab > Run)
2. Export this Live Script as a PDF file (Live Editor > Export)
3. Upload **both the M code file and PDF export** file to Gradescope